Pu Yang

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Education

University of Victoria
Ph.D. in Electrical Computer Engineering (GPA:3.9/4.0) — *May 2020 – Present*
Advisor: Dr. Lin Cai

Northwestern Polytechnical University
M.Sc. in Computer technology (GPA: 3.82/4.0) — *Aug 2017 – May 2020*
Advisor: Dr. Shining Li
B.Eng. in Software Engineering — *Aug 2012 – May 2016*

Research Interests

Trustworthy AI & Security	AI firewalls, secure LLM agents, content safety, and adversarial robustness, including model poisoning and backdoors
Agentic LLM Systems	Safe and reliable tool use, workflow governance, and least-privilege API execution
Efficient Language Models	Small language models (SLMs), KV-cache optimization, and system-level efficiency
Generative Models for Security	Discrete diffusion models for controllable security data synthesis
Distributed & Networked Systems	Federated learning security, 6G architectures, and delay-constrained communication systems

Work Experience

Edison Research Center, Huawei Canada — *Jan. 2026 – Present*
Senior Machine Learning Researcher — *Canada*

- Lead a team of four Ph.D. researchers on AI-for-security projects, including secure agents, AI firewall systems, and security dataset generation
- Develop data synthesis methods using discrete diffusion language models to construct and augment AI security datasets
- Research model poisoning defenses for federated learning to improve robustness against adversarial training and compromised clients

Edison Research Center, Huawei Canada — *Dec. 2024 – Dec. 2025*
Machine Learning Researcher — *Canada*

- Researched AI firewall architectures for detecting, filtering, and mitigating unsafe or policy-violating model interactions
- Built and evaluated AI agent safeguards for content governance, secure workflow execution, and trustworthy response generation
- Investigated small language model adaptation and KV cache strategies for efficient security-oriented inference

6G Network Architecture Design — *Dec. 2020 – Jul. 2023*
Ph.D student — *Victoria, Canada*

- Self-Evolving and Transformative Protocol Architecture for 6G design
- Updatable eXpress Transport Protocol design
- Delay-Guaranteed Routing Protocol design

Computing Research Institute of Aviation Industry Corporation of China — *Dec. 2018 – Dec. 2019*
Researcher — *Xi'an, China*

- Sub-6G (4 200 MHz - 4 400 MHz)channel modeling

- Wireless communication system design based on Universal Software Radio Peripheral (USRP)
- Excellence Research Intern Award Recipient

Talks

Toward the 6G network architecture — *July 2022*

Wireless Networking Technology Forum Huawei Canada, Victoria, Canada

- Presented key design principles and challenges for the development of 6G networks, including ultra-low latency, massive connectivity, and enhanced security. Discussed the role of AI-driven network management and potential use cases such as smart cities and autonomous systems.

Multi-path routing protocol to support Quality of Service — *Sep, 2023*

IEEE MASS 2023, Toronto, Canada

- Discussed methods to improve network stability in case of congestion and network jamming through the application of multi-path routing protocols, emphasizing their ability to maintain service quality under varying traffic conditions.

Deadline Driven Routing Protocol — *May 2024*

IEEE Conference on Computer Communications, Vancouver, Canada

- Presented the Deadline Driven Routing (DDR) protocol, designed to guarantee service with strict delay constraints in real-time communication systems. Discussed the design, implementation, and evaluation of DDR, focusing on its performance in achieving delay guarantees and improving reliability in delay-sensitive applications.

Enhancing Cloud Computing with Networking Technique — *July 2024*

GenAI and Future Network Forum (Ericsson), Victoria, Canada

- Explored the integration of advanced networking techniques to enhance the efficiency and scalability of cloud computing infrastructures. Demonstrated how novel routing protocols and optimization strategies can improve cloud service delivery, particularly in large-scale distributed environments.

Professional Service

Reviewer IEEE INFOCOM, IEEE/ACM Transactions on Networking (ToN), Computer Networks, IEEE Internet of Things Journal, and ACM SIGKDD (KDD)

Awards & Scholarships

2024	Chinese Government Award for Outstanding Self-financed Students Abroad
2020–2023	UVic Graduate Fellowship; Excellent Research Intern Achievement
2017–2019	NWPU Graduate Scholarship; ACM-ICPC Second Prize of NWPU (2013)

Selected Publications

- **P. Yang**, T. Chang, L. Cai, "DDR: A Deadline-Driven Routing Protocol for Delay Guaranteed Service" IEEE International Conference on Computer Communications (INFOCOM), Jan 15. 2024.
- **P. Yang**, L. Cai, T. Chang, "DGR: Delay-Guaranteed Routing Protocol" The 20th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), July. 2023.
- X. Ren, L. Cai, **P. Yang**, and J. Ji, "Congestion-aware Delay-guaranteed Scheduling and Routing with Renewal Optimization," Computer Networks, June 2023.
- X. Fan , S. Li , W. Cui , **P. Yang**, G. Huang, "Mixed-Numerology Channel Division for Wireless Avionics Intracommunications", Wireless Communications and Mobile Computing, April. 2022
- **P. Yang**, L. Cai, Z. Huang, J. Pan, "ROMAM : An Intelligent Distributed Routing Protocol Architecture", USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2025 (submitted)